

Part II - Stochastic Simulation

Chapter 3 : Generating and Simulation of Random Variables

Dr. Nawel Arrar Remita
Third Year - ENSIA

October 2025

Introduction

When systems are modeled as stochastic processes or queueing systems, they become complex and dynamic, analytical or numerical solutions may become intractable. In such cases, a computer program that mimics the behavior of the system can be used, the main purpose of which is to study the performance of a complex system.

Computer programming or simulation is carried out with several random values, and the behaviors thus modeled are recorded for analysis.

The aim is to generate a sequence of numbers that will appear to be random.

Generating Random numbers

- Key to good simulation lies in the quality of the random number generators used.
- Computer generated random numbers are actually pseudorandom numbers since they all start with a seed that is not random.
- The numbers in the sequence represent outcomes of uniform random variable.
- We are therefore interested in a source capable of producing a sequence $\{u_i\}$ of independent, uniformly distributed random numbers on $[0, 1]$. These numbers are called **pseudo-random**.
- From such a sequence, it is possible to generate any other sequence of random variables following a specific probability distribution.

Generating Random numbers

An algorithmic generator should have a strong mathematical foundation, be simple, and easy to program. The algorithm should be recursive, in the form:

$$u_{i+1} = f(u_i, u_{i-1}, \dots, u_{i-k}) \text{ with } k < i \text{ and } i \geq 1,$$

or

$$u_{i+1} = f(u_i) \text{ with } i \geq 1.$$

Here f is a given function with values in the range $[0, 1]$.

A good random number generator should have a long period before the random numbers start to repeat.

Congruential Methods

The Linear Congruential (LC) Method is a widely used technique for generating random numbers. In this method, the next random number y_i is generated from the current random number y_{i-1} using the following equation:

$$y_i = (a \times y_{i-1} + b) \bmod m,$$

where a and b are non-negative constants.

To produce m different numbers, the following conditions must be hold:

- m and b are relatively prime.
- Any prime factor of m divides $(a - 1)$.
- Uniformly distributed numbers $\{u_i\}$ on $[0, 1]$ are obtained by normalization as follows: $u_i = \frac{y_i}{m}$, $i \geq 0$.
- The generator is initialized with a seed y_0 . Generally, m is chosen to be a power of 2 or a multiple of 100.

Congruential Methods

To expand the range of generated numbers and reduce correlation between successive numbers, several variations of the LC method have been proposed:

- Multiplicative LC Method $y_{i+1} = ay_i \bmod m$, with a maximum period of $m - 1$.
- Additive LC Method $y_{i+1} = (y_i + y_{i-1}) \bmod m$.
- Mixed LC Method $y_{i+1} = (ay_i + b) \bmod m$, where $0 < a < m$ and $0 < b < m$. The maximum period is m .

In these methods, a is called the multiplier, b the increment, m the modulus.

Von Neumann's Middle-Square Mthode

This is one of the earliest methods for generating random numbers.

Algorithm

- 1 Choose an initial seed u_0 ;
- 2 Calculate $a = u_0^2$;
- 3 Set u_1 as the middle digits of the decimal part of a ;
- 4 Set $a = u_1^2$;
- 5 Repeat from step 3.

Remark

At each iteration, the number must retain as many significant digits as the initial seed u_0 .

Main Drawback: If at any step one of the numbers becomes zero, the entire sequence will consist of zero terms going forward.

Validation of Generators

To verify that the sequence $\{u_i\}$ is truly random and follows a uniform distribution, we need to ensure that as $n \rightarrow \infty$ the empirical mean $\bar{U} = \frac{1}{n} \sum_{i=1}^n u_i$ approaches $\frac{1}{2}$. In addition to this, one of the following criteria must be satisfied:

Criterion 1: Let the variance be defined as

$$\text{Var}(U) = \frac{1}{n} \sum_{i=1}^n (u_i - \bar{U})^2. \text{ Then}$$

$$\text{Var}(U) \rightarrow \frac{1}{12} \text{ (convergence in probability).}$$

Validation of generators

Criterion 2: We divide the interval $[0, 1]$ into m subintervals

$I_j = \left[\frac{j-1}{m}, \frac{j}{m} \right]$, such that $1 \leq j \leq m$ and $m \ll n$.

Let V_j be the number of points u_i that fall into the interval I_j and

$n = \sum_{j=1}^m V_j$ represent the total number of observations. Then, we

have:

$$\frac{V_j}{n} = \frac{1}{m}, \forall 1 \leq j \leq m.$$

Validation of generators

Criterion 3: Chi-Squared Uniformity Test

Consider the sequence $(I_j)_{1 \leq j \leq m}$ as a partition of the support of the variable being tested. Let the null hypothesis $\mathcal{H}_0$: be that the sequence $\{u_i\}$ i.i.d and uniformly distributed in $\mathcal{U}[0, 1]$.

We consider $p_j = \mathbf{P}(U \in I_j) = \frac{\text{longueur } I_j}{1-0}$, such that $\sum_{j=1}^m p_j = 1$.

We calculate $W = \sum_{j=1}^m \frac{(V_j - np_j)^2}{np_j} = \chi_r^2$ follows a Chi-Squared

distribution with $r = m - 1$ degrees of freedom. If $\chi_{m-1}^2 > \chi_{\alpha, m-1}^2$, where $\mathbf{P}(\chi_{m-1}^2 > \chi_{\alpha, m-1}^2) = \alpha$ and $\alpha - 1$ is the confidence level (with $\alpha = 0.01$ or 0.05), we reject $\mathcal{H}_0$.

Validation of generators

Criterion 4: If $U \in \mathcal{U}[0, 1]$, then

$$E[U] \pm \sigma_U = \frac{1}{2} \pm \frac{1}{2\sqrt{3}}$$

and the proportion of points in the interval $[0.21132; 0.78868]$ should be close to $2\sigma_U = 0.57736$.

Validation of generators

Criterion 5: (Kolmogorov-Smirnov Test)

This test is used when the distribution being tested has a continuous cumulative distribution function $F(x)$. The test statistic is defined as

$$D_n = \max |F_n(x) - F(x)|,$$

where $F_n(x)$ is the empirical cumulative distribution function.

Let $\{u_i\}$ be the observations, and we state the null hypothesis

$\mathcal{H}_0 : \{u_i\} \text{ i.i.d } \in \mathcal{U}[0, 1]$.

1. Order the observations in ascending order: $u_1 \leq u_2 \leq \dots \leq u_n$
2. Calculate $D^+ = \max \left| \frac{i}{n} - u_i \right|$ and $D^- = \max \left| u_i - \frac{i-1}{n} \right|$.
3. Calculate $D = \max \{D^+, D^-\}$.
4. The critical value D_α is read from the table for a specified α and given sample size n .
5. If $D_\alpha > D$, then $\mathcal{H}_0$ is accepted at the α significance level.

General Procedures for Simulating Random Variables

To generate a sequence $\{x_i\}$ from the probability law of a random variable X with cumulative distribution function $F(x)$.

We aim to create the sequence $\{x_i\}$ from a prior sequence of random numbers $\{u_i\}$ drawn from a uniform distribution on $[0, 1]$. While there are methods based on the specific distribution we wish to generate, there are also several approaches that exploit the unique properties of the distribution itself.

Inverse Method for Continuous Case

Theorem

Let F_X be a cdf defined on the interval $[a, b]$. The quantile function is defined as:

$$F_X^{-1}(u) = \inf\{x \in [a, b] : F_X(x) \geq u\} \text{ where } 0 \leq u \leq 1.$$

The random variable $X = F^{-1}(U)$ has the cumulative distribution function F_X when U follows a uniform distribution on $(0, 1)$.

Algorithm

- 1 Generate $u \in U(0, 1)$.
- 2 Calculate $F_X^{-1}(u)$
- 3 Return $X = F_X^{-1}(u)$.

Inverse Method for Discrete Case

In the case where the continuity constraint is no longer imposed on the function F , we use

$$F^*(u) = \min \{x : F(x) \geq u\}$$

instead F^{-1} . So, if $u \in U[0, 1]$, then the condition $F(x_{k-1}) < u \leq F(x_k)$ implies $X = x_k$, for all $k \in \{1, 2, \dots, n\}$.

Algorithme

- ① Arrange the values $x_1, x_2, \dots, x_n$ such that $x_1 < x_2 < \dots < x_n$.
- ② Generate a random value $u \in U[0, 1]$.
- ③ Determine the output based on the following conditions:
 - If $0 < u \leq p_1$, then output x_1 .
 - If $p_1 < u \leq p_1 + p_2$, then output x_2 .
 - If $p_1 + p_2 < u \leq p_1 + p_2 + p_3$, then output x_3 .
 - If $\sum_{i=1}^{k-1} p_i < u \leq \sum_{i=1}^k p_i$, then output x_k , where $0 \leq k \leq n$.

Accept-Reject Method

Principal of the Method

The Inverse-Transform method required knowing the cumulative distribution function, $F_X(.)$.

However, there are many cases where we do not know the c.d.f., but only know the p.d.f. $f_X(x)$. Thus we need a new method.

This method can be used if the density of the random variable to be generated $f_X(x)$ is bounded and x belongs to a bounded domain, that is $a \leq x \leq b$.

It can be summarized in four steps:

Accept-Reject Method

Principal of the Method

- 1 Normalize the domain of f using a scale C , as follows:

$$0 \leq g(x) = \frac{1}{C} f(x) \leq 1, \forall x \in [a, b];$$

- 2 Define x as a linear function at point $u \in \mathcal{U}[0, 1]$ then
 $x = a + (b - a) u;$
- 3 Generate a pair of numbers $u_1, u_2 \in \mathcal{U}[0, 1];$
- 4 Test: If $u_2 \leq \frac{1}{C} f(a + (b - a) u_1)$ accept $x = a + (b - a) u_1$, otherwise, reject and choose another pair.

Accept-Reject Method

Principal of the Method

Algorithme

- 1 Generate $u_1 \in \mathcal{U}[0, 1]$;
- 2 Calculate $g(u_1) = \frac{1}{c} f(a + (b - a) u_1)$;
- 3 Generate $u_2 \in \mathcal{U}[0, 1]$;
- 4 Compare u_2 à $g(u_1)$; if $u_2 \leq g(u_1)$, accept $x = a + (b - a) u_1$, otherwise, reject (u_1, u_2) ;
- 5 Return to step 1.

Accept-Reject Method

Generalized Method (Generalized Von Neumann)

We aim to bound the density $f_X(x) > 0$ using $g(x) > 0$ an instrumental density of a random variable Y , and the scaling factor C such that $f_X(x) \leq Cg_Y(x), \forall x \in [a, b]$. It is important that $C \geq 1$.

Algorithm

- 1 Find continuous r.v. Y s.t. $g(x) > 0$, and let C be a constant s.t. $\frac{f_X(x)}{g(x)} \leq C, \forall x$ s.t. $f_X(x) > 0$.
- 2 Generate $u_1 \in \mathcal{U}[0, 1]$.
- 3 Generate an instance x of Y .
- 4 Generate $u_2 \in \mathcal{U}[0, 1]$.
- 5 Compare u_2 to $\frac{f_X(x)}{Cg(x)}$; if $u_2 \leq \frac{f_X(x)}{Cg(x)}$, accept $X = x$ and stop. Else reject u_1 and return to step 2.

Composition Method

This method assumes that a probability density $f(x)$ can be expressed as a mixture of probabilities and density functions $g_i(x)$ such that

$$f(x) = \sum_{i=1}^n p_i g_i(x) \quad (9)$$

where $\{p_i\}$ are the probabilities for a discrete random variable N taking values in $\{1, 2, \dots, n\}$ such that $p_i = \mathbb{P}(N = i)$, $0 \leq p_i \leq 1$, $\sum_{i=1}^n p_i = 1$; and $g_i(x)$ is the probability density function of a continuous random variable X_i , $i \in \{1, 2, \dots, n\}$

Composition Method

Algorithme

- 1 Generate $u_1 \in \mathcal{U}[0, 1]$;
- 2 Generate the value of N from the distribution $\{p_i\}$ setting $N = j$ if $\sum_{i=1}^{j-1} p_i < u_1 \leq \sum_{i=1}^j p_i$;
- 3 Generate $u_2 \in \mathcal{U}[0, 1]$;
- 4 Generate a value of X_j (if $N = j$) from the distribution with density $g_N(x) = g_j(x)$ using an appropriate algorithm (inversion or other).

$$X_N = X_j = F_j^{-1}(u_2), \text{ where } F_j(x) = \int_{-\infty}^x g_j(y) dy$$

- 5 $X = X_j$;